Lab 4

(20+10+10 = 40 points total)

1. Write an assembly program **using only conditional instructions** which is equivalent to the following C code (of course, you don't need to code the `printf` statement):

```c
#include <stdio.h>

int main()
{
    unsigned int x = 7;
    unsigned int y = 23;
    if(x > y){
        x = x - y;
    }
    else if(y > x){
        y = y - x;
    }
    printf("x = %d, y = %d\n", x, y);
    return 0;
}
```

2. Define the two arrays `x` and `y` as shown below. Then, write an assembly program using a series of LDR and STR instructions such that the array `y` is equal to [5, 4, 3, 2, 1] (i.e., the array `x` reversed) after program completion. **Do not use a loop/branches** (other than the `B end` already shown). **Note**: You can use at most three registers, R0, R1 and R2.

```
.global _start
_start:

    Your code here

    B    end

_data:
x:   .word 1,2,3,4,5
y:   .word 0,0,0,0,0

_end:
```

3. Define the array `x` as shown below. Then, write an assembly program using a series of LDR and STR instructions such that the array `x` is equal to [5, 4, 3, 2, 1] after program completion. **Do not use a loop/branches** (other than the `B end` already shown). **You cannot use any other array other than x. Array reversal must be achieved using only LDR and STR instructions**. **Note**: You can use at most three registers, R0, R1 and R2.

```
.global _start
_start:

        Your code here

    B    end

_data:
x:    .word 1,2,3,4,5
```

Note: You can work on the lab assignment during the week and just demo it during the lab section next week (the week of May 18). Or you can choose to work on it during the lab hours next week. If you choose the latter option, please note that no grace period will be provided to finish your lab.